TECHNICAL CHANGE REPORT

DragonBox Pyra Kernel 5.6.19

Cortex-M4 / IPU integration, AMMU safety controls, PRM reset corrections and platform stability fixes

BASELINE	5.6.19-pyradef · Debian packaging revision -10
MODIFIED BUILD	5.6.19-pyradef-m4test-m4clean · Pyra M4 v6
SCOPE	15 changed source paths · 14 modified · 1 added
PLATFORM	DragonBox Pyra V5.3 · TI OMAP5432 / ARMv7
SOURCE BASIS	Integrated source patch and verified changed-file manifest
ISSUED	26 October 2026

DOCUMENT PURPOSE

A structured, file-by-file account of all source changes introduced by the experimental M4 kernel project. Installed boot artifacts and explicit non-changes are documented separately so the scope remains auditable.

VERIFICATION BOUNDARY

The source inventory is grounded in the integrated project patch and manifest. A successful source comparison or kernel build does not, on its own, establish successful M4 firmware execution or validate all Pyra hardware functions.

01 Executive overview

The modified kernel introduces a guarded remoteproc path for the OMAP5432 IPU Cortex-M4 subsystem while preserving the original Pyra Linux environment and its PowerVR graphics stack.

Change areas

- **M4 enablement:** Device Tree-aware remoteproc driver, separate M4-enabled DTB, mailbox and remoteproc configuration.
- **CPU0-only safety:** Explicit reset sequencing, AMMU Small Page 0 mapping, readback, rollback and one-start-per-boot guard.
- **PRM / Letux:** Selected Letux 7.3 reset-deassert sequencing fixes and opt-in IPU reset diagnostics/quirk.
- **Platform fixes:** MMC timeout overflow, charger property parsing, TILER allocation errors and DWC3 USB/OTG handling.
- **Build and isolation:** ARM compiler-flag ordering, SELinux export header compatibility, independent kernel release and modules.

Source change matrix

#	Kernel source path	Change
01	arch/arm/Makefile	ARM kernel build flags
02	arch/arm/boot/dts/Makefile	Additional DTB target
03	arch/arm/boot/dts/omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dts	New M4-enabled Pyra DTS
04	arch/arm/boot/dts/omap5.dtsi	OMAP5 IPU node
05	arch/arm/configs/pyra_defconfig	Remoteproc/mailbox config
06	drivers/gpu/drm/omapdrm/omap_dmm_tiler.c	ENOMEM error propagation
07	drivers/iommu/omap-iommu.c	Header dependencies
08	drivers/mmc/host/omap_hsmmc.c	Timeout arithmetic
09	drivers/power/supply/bq2429x_charger.c	DT property conditionals
10	drivers/remoteproc/Kconfig	OMAP5 help text
11	drivers/remoteproc/omap_remoteproc.c	Remoteproc, CPU0, AMMU
12	drivers/remoteproc/omap_remoteproc.h	Mailbox message IDs
13	drivers/soc/ti/omap_prm.c	PRM reset sequence
14	drivers/usb/dwc3/dwc3-omap.c	USB/OTG regulator state
15	security/selinux/include/classmap.h	Exported-header guard

02 Device Tree and kernel configuration

These changes introduce an opt-in IPU description and build target without modifying the original Pyra board DTS.

02 arch/arm/boot/dts/Makefile

- Adds omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dtb to the CONFIG_SOC_OMAP5 Device Tree build targets.
- Retains the original Pyra V5.3 DTB target.

03 arch/arm/boot/dts/omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dts

- NEW source file (10 lines).
- Includes the original V5.3 board DTS.
- Overrides only the IPU node with: &ipu { status = "okay"; };
- The original board DTS remains unmodified.

04 arch/arm/boot/dts/omap5.dtsi

- Adds an IPU node at physical address 0x55020000, initially disabled.
- Compatible string: "ti,omap5-ipu".
- Internal L2 SRAM region: 0x55020000, length 0x10000 (64 KiB); register name "l2ram".
- Connects the IPU IOMMU (&mmu_ipu), two reset controls (&prm_core 0 and &prm_core 1), IPU clock and OMAP mailbox.
- Firmware name: "omap5-ipu-fw.xem4".
- No reserved-memory node or memory-region reference is added.

05 arch/arm/configs/pyra_defconfig

- Changes CONFIG_LOCALVERSION from "-pyradef" to "-pyradef-m4test".
- Enables CONFIG_OMAP2PLUS_MBOX=y.
- Enables CONFIG_REMOTEPROC=y.
- Enables CONFIG_OMAP_REMOTEPROC=y.
- Other defconfig lines are unchanged by the combined source patch.
- The actual distributed build additionally supplies LOCALVERSION=-m4clean, producing the release string 5.6.19-pyradef-m4test-m4clean.

03 Remoteproc and M4 execution controls

The OMAP5 IPU is represented as one Linux remoteproc device. The experimental start path intentionally targets CPU0 only; CPU1 remains held in reset.

10 drivers/remoteproc/Kconfig

- Updates the OMAP remoteproc help text to mention OMAP4 and OMAP5 Device Tree-based IPU support instead of claiming OMAP4-only support.
- The actual feature selection is made in pyra_defconfig.

11 drivers/remoteproc/omap_remoteproc.c

- Replaces the earlier platform-data-oriented OMAP driver with a Device Tree-aware driver based on the Linux 5.7 implementation, adapted to Linux 5.6 interfaces.
- Uses Device Tree compatible matches, firmware-name, reset controls, clocks, mailbox channel and optional timer resources.

- Maps internal remote-processor memories and implements the remoteproc device-address-to-host-address translation callback.
- Supports firmware loading, start/stop, mailbox signaling, watchdog/timer integration, runtime PM and suspend/resume paths where applicable.
- Sets rproc->auto_boot = false; the M4 firmware is not started automatically when remoteproc registers.
- Adapts timer set_load() to the three-argument kernel API, da_to_va() to an int length argument, and boot-address logging to a 32-bit 0x%08x format.
- Adds the built-in module parameter omap_remoteproc.pyra_cpu0_only=1 for the OMAP5 IPU. The parameter defaults to disabled.
- When CPU0-only mode is requested, obtains two independent reset controls, asserts both, and verifies both are held at probe.
- On a firmware start, reasserts CPU0 and CPU1 reset, performs AMMU preflight, releases CPU0 only, and verifies the expected post-start reset state (CPU0 released, CPU1 held).
- Refuses an OMAP5 IPU firmware start if CPU0-only mode is not active; this is an experimental safety constraint, not generic dual-core IPU support.
- On stop, asserts CPU0 and ensures CPU1 remains held.
- Adds an atomic one-attempt-per-kernel-boot guard. A second firmware-start attempt is rejected; reboot is required.
- Implements a guarded AMMU Small Page 0 preparation:
  - Requires CPU0-only mode and both CPU reset lines asserted.
  - Checks RM_IPU_RSTCTRL at 0x4AE06910 equals 0x00000003 before allowing the transition.
  - Reads IPU configuration registers through the host-visible 0x55080000 region, including Page 0 and Page 1 address, translation and policy registers.
  - Accepts only known Page 0 values (ADDR=0, XLTE=0 or 0x55020000, POLICY=0x0b) and known Page 1 values (ADDR=0x40000000, XLTE=0x55080000, POLICY=0x09), together with expected MMU/cache-maintenance state.
  - If necessary, writes 0x55020000 to the Page 0 XLTE register at host physical address 0x550809A0, mapping the M4-local address 0 to the internal IPU L2 SRAM.
  - Reads back the written value and rechecks relevant registers and CPU reset states before CPU0 release.
  - On failed verification, attempts to restore the original XLTE value, keeps CPU0 in reset, and reports an error.
  - Does not write external Linux IOMMU page tables, DDR, CPU1 reset from the AMMU helper, or cache maintenance registers.
- Retains the Letux PRM changes implemented separately in drivers/soc/ti/omap_prm.c; the AMMU helper does not replace the PRM reset driver.
- Removes historical experimental diagnostic branches and their boot-argument switches from the final clean driver.

12 drivers/remoteproc/omap_remoteproc.h

- Updates the source header/SPDX metadata and copyright years.
- Adds mailbox message identifiers for automatic/system suspend, suspend acknowledgement/cancellation, and end-of-message marker: RP_MBOX_SUSPEND_AUTO = 0xFFFFFF10 RP_MBOX_SUSPEND_SYSTEM = 0xFFFFFF11 RP_MBOX_SUSPEND_ACK = 0xFFFFFF12 RP_MBOX_SUSPEND_CANCEL = 0xFFFFFF13 RP_MBOX_END_MSG = 0xFFFFFF14

CRITICAL AMMU MAPPING

The controlled mapping is M4 local address 0x00000000 → physical IPU L2 SRAM 0x55020000, using the Small Page 0 XLTE register at 0x550809A0. The code checks reset states and known register patterns before CPU0 release.

04 Reset management and platform stability

13 drivers/soc/ti/omap_prm.c

- Adds the module parameter omap_prm.pyra_ipu_prm_trace, disabled by default, for read-only IPU CORE PRM snapshots.
- Adds the module parameter omap_prm.pyra_ipu_prm_assert_quirk, disabled by default, for an explicitly requested IPU CORE reset assertion during PRM probe.
- Restricts the optional quirk to the IPU CORE PRM instance (base 0x4AE06700) and rst_map_012.
- Preserves unrelated RSTCTRL bits via read-modify-write, followed by a readback to flush the posted write.
- In reset deassertion, returns early if the requested reset is already released.
- After clearing a reset bit, uses readl_relaxed_poll_timeout_atomic() to wait for RSTCTRL to reflect the cleared state.
- If a reset status register exists, additionally waits for the corresponding RSTST bit using atomic polling.
- These are retained selected Letux 7.3 reset-sequencing corrections; they are NOT a wholesale import of Letux 7.3.
- Optional tracing logs PWRSTCTRL, PWRSTST, RSTCTRL and RSTST before/after reset deassertion and at relevant probe points.

06 drivers/gpu/drm/omapdrm/omap_dmm_tiler.c

- Sets ret = -ENOMEM if the TILER refill DMA allocation fails, before entering the existing error cleanup path.
- Fixes the returned error code; it does not change the PowerVR rendering pipeline.

07 drivers/iommu/omap-iommu.c

- Adds three header includes: linux/atomic.h, linux/moduleparam.h, and linux/string.h.
- No executable IOMMU function or mapping logic is changed by the final combined patch in this file.
- Earlier diagnostic IOMMU experiments are NOT part of this clean source snapshot.

08 drivers/mmc/host/omap_hsmmc.c

- Casts busy_timeout to u64 before multiplying by NSEC_PER_MSEC.
- Prevents 32-bit intermediate arithmetic overflow in the busy response timeout calculation.

09 drivers/power/supply/bq2429x_charger.c

- Removes accidental trailing semicolons from two of_property_read_u32() conditionals.
- Applies precharge-current and charge-termination-current values only when their optional Device Tree properties were read successfully, rather than potentially using an uninitialized local value.

14 drivers/usb/dwc3/dwc3-omap.c

- Checks whether the VBUS regulator is enabled before disabling it when the USB ID signal floats.
- Initializes both USB and USB_HOST extcon states, including explicit OFF/FLOAT states when no cable is present.
- Uses devm_regulator_get_optional() for VBUS: tolerates -ENODEV but preserves other errors, including probe deferral.
- Depopulates child devices on an IRQ setup failure before unwinding runtime PM.

05 Build and exported-header compatibility

01 arch/arm/Makefile

- Moves -msoft-float into the earlier ARM KBUILD_CFLAGS definition, so compiler feature probes and ordinary kernel C compilation both use the correct no-hardware-FP kernel ABI.
- Removes the duplicate later -msoft-float occurrence.
- This is a build-flag ordering fix, NOT an enablement of NEON in the kernel and NOT a change to the Pyra userspace ABI.

15 security/selinux/include/classmap.h

- Restricts the PF_MAX compile-time address-family check to builds where __EXPORTED_HEADERS__ is not defined.
- This is a header/export build-compatibility change.

06 Built artifacts and boot integration

The following differences are produced by the build and installation workflow. They are not additional source-file modifications.

RELEASE	Original: 5.6.19-pyradef | New: 5.6.19-pyradef-m4test-m4clean
MODULES	/lib/modules/5.6.19-pyradef-m4test-m4clean/ (946 .ko files in supplied staging archive)
KERNEL IMAGE	/boot/vmlinuz-5.6.19-pyradef-m4test-m4clean
M4 DEVICE TREE	Additional M4-enabled DTB, installed as ...-m4clean.dtb; generated from the -m4test.dts source
INITRAMFS	Target-generated /boot/initrd.img-5.6.19-pyradef-m4test-m4clean, where the boot entry requires it
BOOT ENTRY	LABEL pyra-m4clean in /boot/extlinux/user.cfg
BOOT ARGUMENT	omap_remoteproc.pyra_cpu0_only=1

PowerVR compatibility

The combined patch does not modify the original PowerVR SGX driver source or native EGL/GLES2 userspace libraries. The staged module tree includes the DDK 1.14.3699939 pvrsrvkm_omap5_sgx544_116.ko module built for the new kernel release. Hardware compatibility still requires on-device testing.

Recovery and default boot

The original extlinux entries l0, l0r and l0h remain available. In the user-shared configuration, extlinux.conf contained both “default pyra-m4clean” and “default l0”, so the active kernel remained 5.6.19-pyradef. That configuration conflict is not a source change in the modified kernel.

Separate M4 firmware

The test firmware-v12 project is a separate Cortex-M4 executable, not a kernel source modification and not part of the kernel installer. The Device Tree firmware-name is omap5-ipu-fw.xem4.

07 Explicit exclusions and verification limits

- The original Pyra V5.3 board DTS is unchanged; a separate M4-enabled DTS is added.
- No modifications to the PowerVR SGX driver source or the native userspace graphics stack.
- No Mesa, GL4ES or GLX replacement is introduced.
- No general-purpose DDR AMMU mapping, reserved-memory/CMA allocation or RPMsg application is introduced.
- No routine CPU1 startup is enabled in CPU0-only mode.
- No Phase 9.13+ mailbox IPC feature is included in this v6 snapshot.
- Only selected PRM reset fixes are carried over; this is not a complete Letux 7.3 kernel port.
- Source consistency does not establish successful M4 firmware execution or stable PowerVR, USB, MMC/F2FS, suspend or charging operation.
- The earlier reported uname -r value was 5.6.19-pyradef; activation of the new kernel was not independently established in that output.

08 Provenance and integrity record

The authoritative comparison baseline is the original Pyra kernel source with the Debian packaging revision -10 diff applied. The integrated source distribution was previously checked against that baseline with zero patch fuzz; the 15 resulting source files matched its manifest byte-for-byte.

Baseline archives

Original tarball: `linux-5.6.19-pyradef_5.6.19-pyradef.orig.tar.gz`

Original SHA-256: `4ca8561817b9bae8b462b5fa67a6b38a9f77d50c0661bc5f00b8d795f208ba6d`

Debian source diff: `linux-5.6.19-pyradef_5.6.19-pyradef-10.diff.gz`

Diff SHA-256: `28533e0045f53eda551ea48bd6c22570905f5b26ea0c4cef432a224d308a8270`

Source file SHA-256 fingerprints

Each pair records the SHA-256 of the baseline file and the modified file. “New file” denotes the newly added board DTS. Full 64-character digests are included for reproducibility.

01 arch/arm/Makefile
BASE `7f364f93fee2324ad9a2c92bcfcd26424b68c43e0d0ea8dd2aa888a206a93800`
NEW `b0011d8bf734900ef258cfe9d1a8ba12d13e66a9cbf2cf5638b1b1709659bedd`

02 arch/arm/boot/dts/Makefile
BASE `208b0ff26e17fabbbf5add7af51d20744ed5c5566528d792d2d8f0636540c956`
NEW `293b0ee9fa34b9ec69ba52cabe3f90dacefb0194d91824c2a545c048d9e35ec5`

03 arch/arm/boot/dts/omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dts
BASE `(new file)`
NEW `48855d76ae2428bad78bad97ca8eae8f50fd815ab20e8b0c161f70ff6a064da1`

04 arch/arm/boot/dts/omap5.dtsi
BASE `0cc541af315cbff678b0ed7821ca1807def7009e03f39f56a3c37a6ceff725e9`
NEW `57192bac9b668f22205fc1c7296528e127ffa4e43c0d349675dd4d60a5d2dbc5`

05 arch/arm/configs/pyra_defconfig
BASE `ad7e6ebc382d1def0de4f32f4850b696061af8b302b244a9bb4c3e091151d829`
NEW `d7ca694bdc3584957c062b24230936d7b64e66b6d2fbf3c4f3d1650c5eede6bb`

06 drivers/gpu/drm/omapdrm/omap_dmm_tiler.c
BASE 8c16917a90eb8dd61d03a40e0fb383086a25ca6fc41d45bc86a5cd30f6c2b87d
NEW 4482e2b0fb8334547c4a2b2f09c4507d9857cb4140b83a28ac776228cbde4b6e

07 drivers/iommu/omap-iommu.c
BASE 1eac9415a73844bb686e0461ca0015e290be399f8ed5bc2b7769ae40d3e9dbbb
NEW 12adf92a7718d6945f9aa6c1e9701afa0e2e434bd4c7fb5f5bd19f764a513495

08 drivers/mmc/host/omap_hsmmc.c
BASE a65a0d553d14a446aa8c3647c6e3754873c2dbaa7f9779a88831e1d985f63b1c
NEW 20ae9ee02af9dfaab27f1b8c2b084e3baa3e9ab5a68e293c4dd164b533adb7a7

09 drivers/power/supply/bq2429x_charger.c
BASE 5a64958e773fe39bb0233d81f52106067087b07b0d9eb93e516037683199ffdd
NEW 02ccedb014c411ab6742b1fd8469601e7a9b51c5832d545044ac723492dcd2c9

10 drivers/remoteproc/Kconfig
BASE f9a97c433d85aecdb0227ec219fe1b7cceee818bf7644c92633beaf2f8c4f4ec
NEW a4ae88077ba0ddfd250dccfb0eb95d1d9a5606fdedaca1b0e388efdcc2a1888f

11 drivers/remoteproc/omap_remoteproc.c
BASE 0bc8dc2bf5334553767d3e1360813f8d164e32b5e4302defd4890fbcb9aa8869
NEW 28bf710d654cd99e048cdbd2a89078b20c44804b952fa3f106cba6c34d3baa06

12 drivers/remoteproc/omap_remoteproc.h
BASE 36e71c0cf1941819c160dfa03aeb4dda4666110eb194cad298166de3adf7df26
NEW 5b592ec8acf6e0d42b426606b4b4262f596970c2e86d10038d54900db6265400

13 drivers/soc/ti/omap_prm.c
BASE cceb0f3b5a02feb8ee70f9e74c625660e11b59e0ef2267b8eb380b23a463d813
NEW e49a610d32c1089d8e7531f19573dcb6249276d864d1d567b028d83465646a5a

14 drivers/usb/dwc3/dwc3-omap.c
BASE 464b4239af35d9ccb06576e0fa99bfd6744c481a66280fa87abbfad011ed2f12
NEW 1bc0e7309da430285d62db7a6835122f27a8e21f23f0401abff78fb35ae0e7b6

15 security/selinux/include/classmap.h
BASE 8eb03ae562a9dfb0ad0b4927e1af50b1177909e20fecc082128ecce3be22c6a6
NEW 7d454deadeb350b92ac490656fce5f95d86a1d9080ee6a61ba473c60a1d72c4b

Complete line-by-line source patch

For an exact line-by-line representation of every source change, see the companion text artifact “Pyra_Kernel_5.6.19_pyradef_M4_v6_Complete_Changes_EN.txt”, Appendix A. This report prioritizes a navigable engineering overview while retaining the complete source-file inventory and integrity fingerprints.